

WaferGuard ML

Sprint 1: SECOM Dataset Exploration

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What is SECOM Data?

Sensor data from a semiconductor manufacturing process.

- **590 sensor measurements** per production entity
- **1,567 samples** (wafers/chips)
- **Binary outcome:** Pass (-1) vs Fail (1)
- **Source:** UCI Machine Learning Repository

Real-world sensor data is notoriously “messy” and redundant

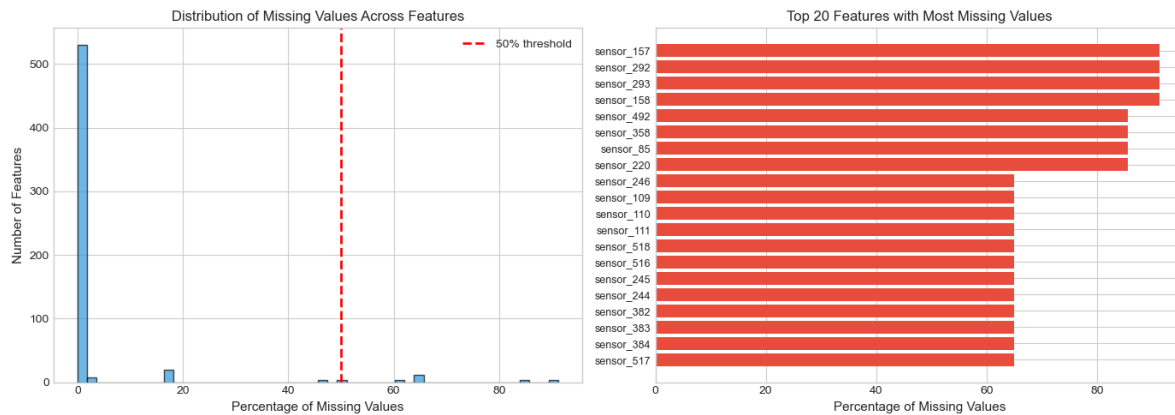
The Challenge

- High-dimensional sensor data (590 features)
- Severe class imbalance (~93% Pass, ~7% Fail)
- Significant missing values across features
- Many redundant/correlated sensors

Dataset Overview

Attribute	Value
Samples	1,567 production entities
Features	590 sensor measurements
Labels	Pass (-1) / Fail (1)
Collection Period	89 days (Jul-Oct 2008)

Missing Value Analysis

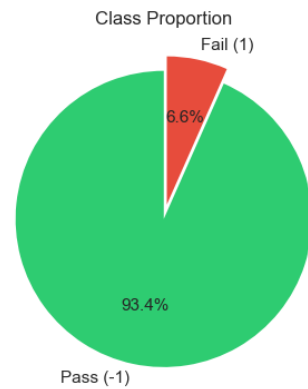
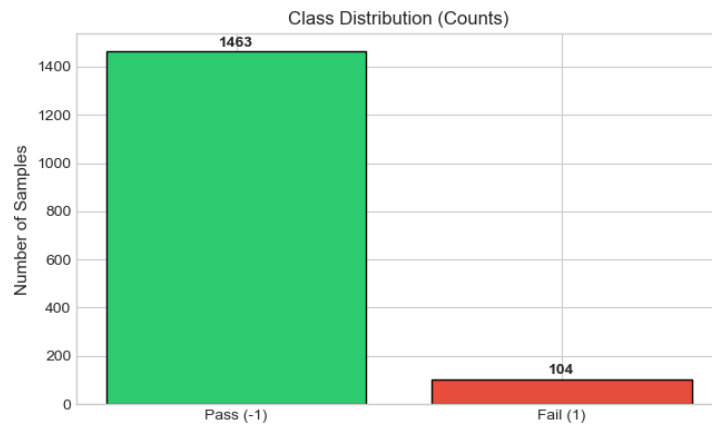


Missing Value Distribution

- **Features with ANY missing:** 538 out of 590 (91%)
- **Features with >50% missing:** 28 features
- **Average missing per sample:** 26.8 features (4.5%)

Strategy: Drop features with >50% missing; impute the rest

Descriptive Statistics

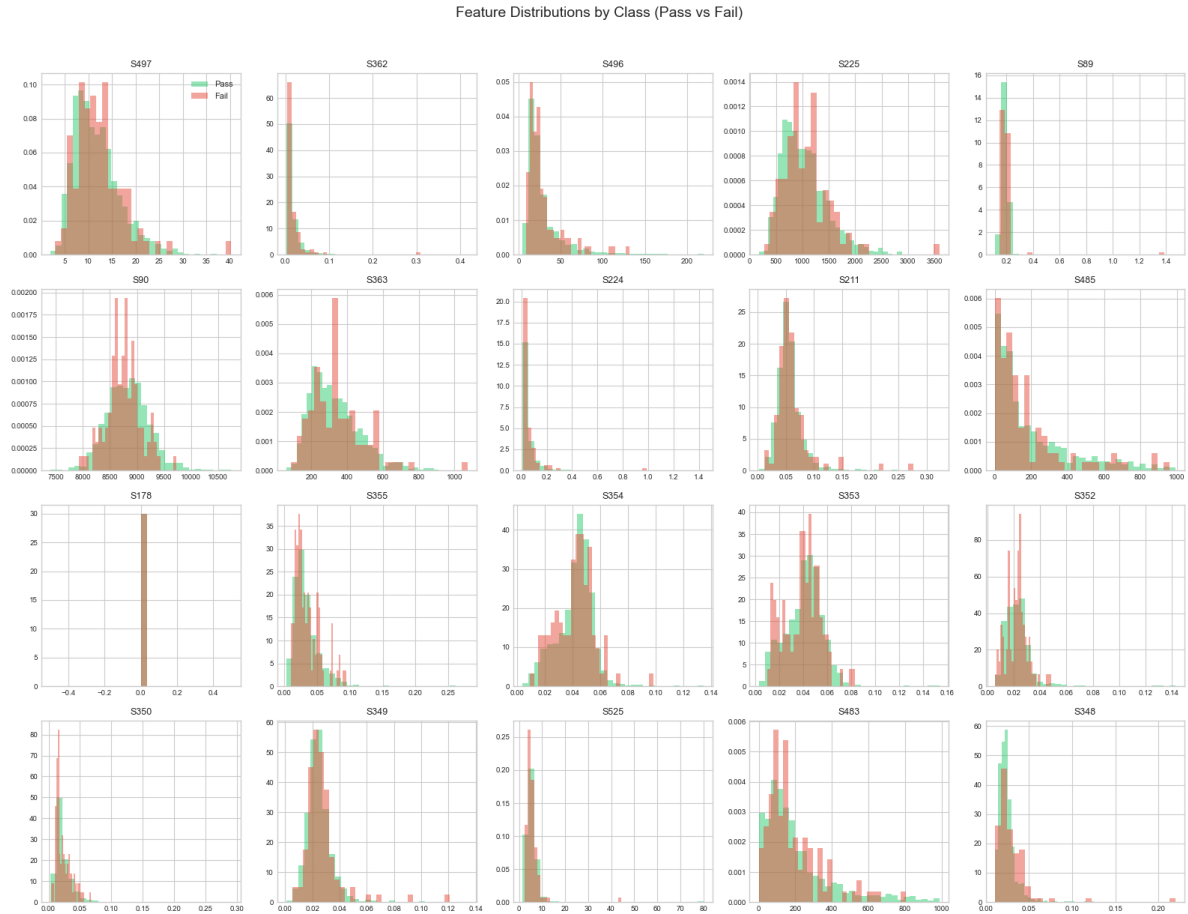


Constant Features

Some sensors show **zero variance** (useless for prediction).

- Example: `sensor_5` = 100.0 for all samples
- These provide no discriminative power
- Must be identified and removed

Class Distribution



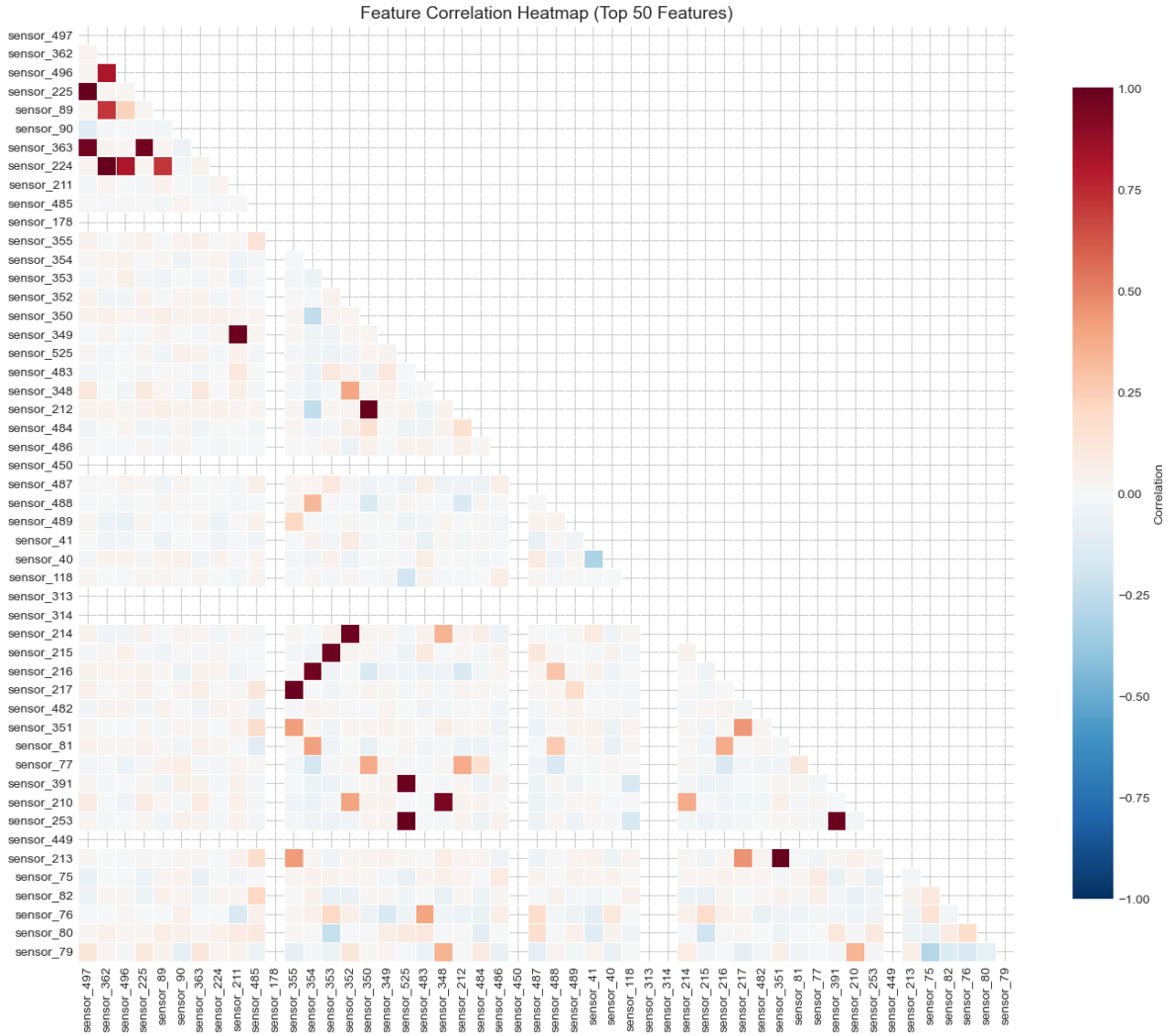
Class Imbalance

The dataset is heavily skewed towards “Pass”.

Class	Label	Frequency
Pass	-1	~93%
Fail	1	~7%

Risk: A model predicting “Pass” everywhere achieves 93% accuracy but fails the objective.

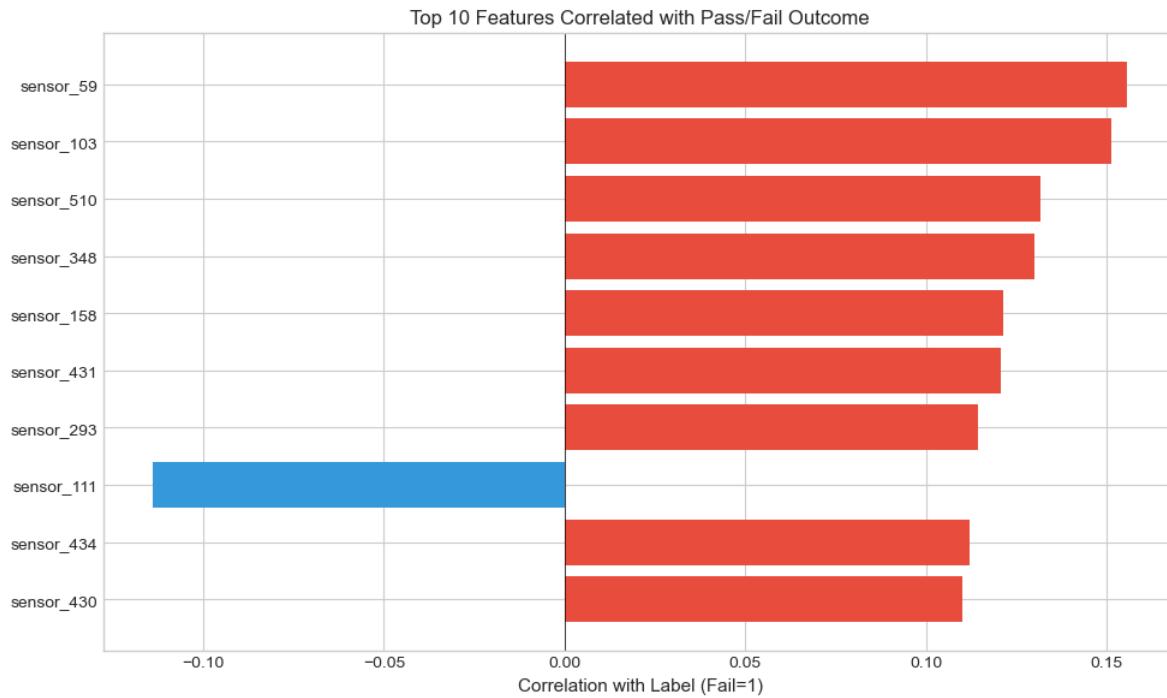
Correlation Analysis



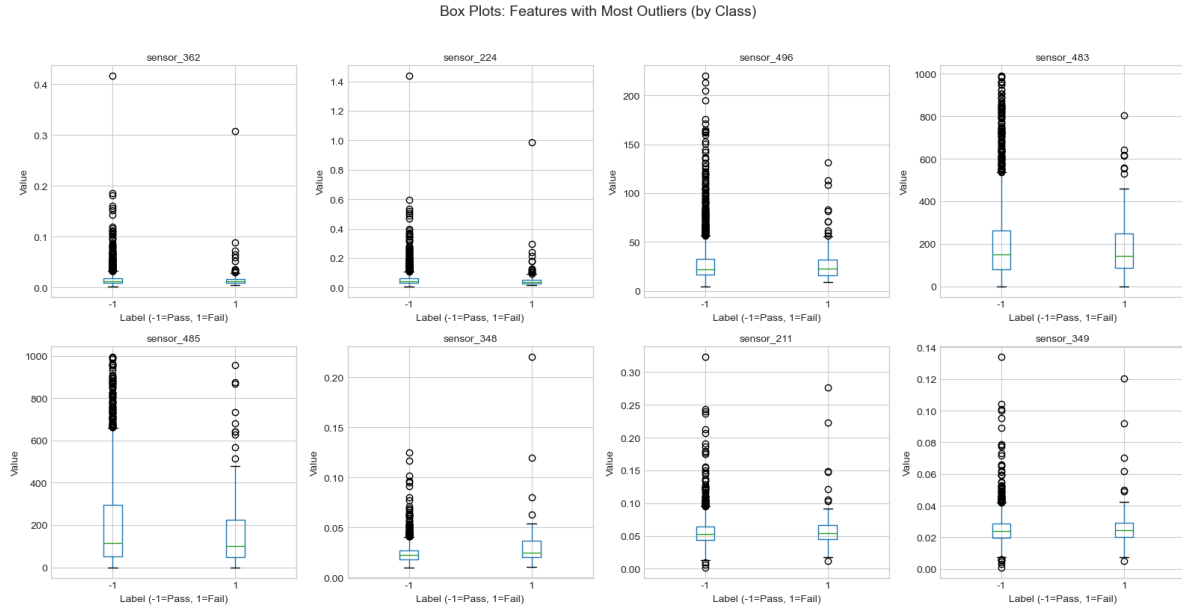
Feature Correlations

- Many sensors are highly correlated
- Redundant features increase model complexity
- Strategy: Remove one of each highly correlated pair ($|r| > 0.95$)

Additional Analysis



Feature Distribution



Preprocessing Strategy

1. Cleaning:

- Remove constant features (Zero Variance)
- Remove high missingness columns ($>50\%$)

2. Feature Reduction:

- Address multicollinearity ($|r| > 0.95$)
- Consider PCA

Handling Imbalance

To enable effective defect detection:

- **Stratified Sampling:** Maintain 7% failure rate in splits
- **SMOTE:** Generate synthetic failure examples
- **Class Weights:** Penalize missing a failure more heavily

Next Steps

1. Execute preprocessing pipeline
2. Baseline Model Training (Logistic Regression / Random Forest)
3. Evaluate using **Recall** and **F1-Score** (Accuracy is misleading here)

Dataset Summary

- 1,567 Production Samples
- 590 Sensor Features
- Binary Pass/Fail Labels
- 89 Days of Production Data

Data Quality Assessment

- Significant Missing Values (91% of features affected)
- Constant Features Present
- Severe Class Imbalance (93/7 split)
- Preprocessing Pipeline Defined
- Ready for Model Development